

Week 7 — Demo: Fourier Decomposition: Amplitude, Phase, and Sampling

Geophysics 3A: Potential Fields & Geothermics

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1 Purpose of the Demonstration

This demonstration is designed to build intuition for what a Fourier transform does and does not do. In particular, it addresses:

- How a signal may be constructed from known frequencies, amplitudes, and phases
- How a discrete Fourier transform (DFT) estimates those parameters from sampled data
- Why amplitude recovery is affected by sampling and windowing
- Why phase recovery depends on reference conventions
- How these ideas carry directly into spatial spectral methods in potential field geophysics

Although the demonstration uses time-domain signals, the mathematics applies identically to spatial data.

2 Signal Construction

We construct a signal as a sum of sinusoidal components:

$$x(t) = \sum_{i=1}^{N_f} A_i \sin(2\pi f_i t + \phi_i), \quad (1)$$

where:

- f_i are known frequencies,
- A_i are amplitudes,
- ϕ_i are phase offsets.

In the demonstration, amplitudes follow a power-law decay,

$$A_i = f_i^{-n}, \quad (2)$$

to mimic the tendency for geophysical spectra to be dominated by long wavelengths.

2.1 Linear Phase and Time Shifts

A linear phase is imposed by choosing

$$\phi_i = -2\pi f_i t_0, \quad (3)$$

which gives

$$x(t) = \sum_i A_i \sin[2\pi f_i (t - t_0)]. \quad (4)$$

Thus, a linear phase spectrum corresponds to a constant time shift t_0 . This is a fundamental Fourier property:

$$x(t - t_0) \iff X(f) e^{-i2\pi f t_0}. \quad (5)$$

The key information carried by phase in this context is therefore the *slope* of phase as a function of frequency, not its absolute value.

3 Discrete Fourier Transform

A discrete Fourier transform of N samples $x_n = x(n\Delta t)$ is defined as

$$X_k = \sum_{n=0}^{N-1} x_n e^{-i2\pi k n / N}, \quad (6)$$

where k indexes discrete frequency bins

$$f_k = \frac{k}{N\Delta t}. \quad (7)$$

The FFT returns complex coefficients X_k ; from these we extract:

$$\text{Amplitude: } |X_k|, \quad (8)$$

$$\text{Phase: } \arg(X_k). \quad (9)$$

3.1 Amplitude Scaling

The DFT does not return amplitudes in the same units used to construct the signal. For a pure sinusoid of amplitude A sampled N times, the magnitude of its FFT coefficient scales as $\sim N/2$.

To recover physical amplitudes from a one-sided spectrum, we apply the deterministic scaling

$$A_k^{(\text{rec})} = \frac{2}{N G_w} |X_k|, \quad (10)$$

where G_w is the coherent gain of the window function. For a Hann window,

$$G_w = \frac{1}{N} \sum_{n=0}^{N-1} w_n \approx 0.5. \quad (11)$$

This correction removes known numerical scaling but does not make the recovery exact; remaining discrepancies arise from finite record length, bin mismatch, and spectral leakage.

4 Phase Conventions and Interpretation

Phase recovery requires particular care.

4.1 Sine versus Cosine Basis

The signal is constructed using sine functions, while the FFT basis corresponds to complex exponentials (or equivalently, cosines and sines). Since

$$\sin(\omega t + \phi) = \cos\left(\omega t + \phi - \frac{\pi}{2}\right), \quad (12)$$

the recovered FFT phase includes a constant offset of $-\pi/2$ relative to the input phase. This offset is frequency-independent and carries no physical meaning.

Time-domain signal	Spatial potential field
t	x
Frequency f	Wavenumber k
Time shift	Lateral offset
Sampling rate	Station spacing
Linear phase	Spatial translation

4.2 Absolute Phase versus Phase Differences

Phase is inherently defined modulo 2π , and an overall constant phase depends on the choice of time origin. In this demonstration:

- Absolute phase values depend on arbitrary reference choices.
- Phase *differences* and *slopes* carry physical information.

This distinction mirrors geophysical practice:

- In magnetotellurics, phase is referenced to the inducing field and is physically interpretable.
- In potential fields and this demo, constant phase offsets correspond to arbitrary spatial or temporal origins.

5 Connection to Spatial Geophysics

The mathematics is identical if time t is replaced by spatial position x :

These ideas underpin upward and downward continuation, reduction to the pole, and spectral depth estimation.

6 Suggested In-Class Demonstration Outline

The following sequence uses the interactive notebook widgets.

Step 1: Frequency Recovery

- Phase type: linear
- Phase shift: $t_0 \approx 0.2$ s
- Amplitude exponent: $n = 1$
- Padding: $\times 4$
- Window: on

Observe that FFT peaks occur near the input frequencies.

Step 2: Amplitude Decay

Increase the amplitude exponent n from 0 to 2 and observe how spectral shape changes while phase trends persist.

Step 3: Linear Phase and Time Shifts

Vary t_0 and observe:

- A linear phase trend
- Increasing slope with increasing shift
- No change in amplitude spectrum

Step 4: Phase Scrambling

Set phase type to “scrambled.” Observe that:

- The time series becomes complex
- The amplitude spectrum is unchanged
- Phase loses coherent structure

Step 5: Padding and Windowing

Toggle padding and windowing to show:

- Improved phase stability with padding
- Reduced leakage with windowing

Emphasize that improvements are numerical, not physical.

7 Key Takeaways

- Fourier transforms reorganize information; they do not interpret it.
- Amplitude and phase are recovered approximately from sampled data.
- Phase slopes carry physical meaning; absolute offsets often do not.
- The same concepts govern spectral methods in potential field geophysics.